

2. Requirement Analysis Phase

Brief Explanation

The **Requirement Analysis Phase** is an important stage in the Software Development Life Cycle (SDLC). In this phase, the requirements of the proposed system are collected, studied, and clearly documented. The main purpose is to understand what the users expect from the system and what functions the system should perform.

During this phase, information is gathered from users, customers, staff, and other stakeholders through interviews, questionnaires, observations, and discussions. The collected requirements are then analyzed to identify the **functional requirements** and **non-functional requirements** of the system.

Functional requirements describe the activities that the system should perform, such as user registration, login, adding records, updating information, searching, and generating reports. Non-functional requirements describe the quality and performance of the system, such as security, reliability, usability, speed, and scalability.

The requirements are checked for correctness, completeness, and feasibility. Any unclear or conflicting requirements are discussed and clarified with the users. After finalizing the requirements, they are documented in a **Software Requirements Specification (SRS)** document. This document acts as a guide for developers during the design and development stages.

Thus, the Requirement Analysis Phase helps in developing a system that meets the actual needs of its users and reduces errors, confusion, and unnecessary changes during later stages of development.

Code – Requirement Analysis Example

```
# Simple User Requirement Example

users = []

def register_user():
    name = input("Enter your name: ")
    email = input("Enter your email: ")

    user = {
        "name": name,
        "email": email
    }

    users.append(user)
    print("User registered successfully!")

def display_users():
    print("\nRegistered Users:")

    for user in users:
        print("Name :", user["name"])
        print("Email:", user["email"])
        print("-----")

# Main Program
while True:
    print("\n1. Register User")
    print("2. Display Users")
    print("3. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        register_user()

    elif choice == "2":
        display_users()

    elif choice == "3":
        print("Program ended.")
        break

    else:
        print("Invalid choice!")
```

Code Explanation

1. **users = []** – Creates an empty list to store user information.
2. **register_user()** – Collects the user's name and email address.
3. **input()** – Gets required information from the user.
4. **Dictionary** – Stores the name and email together as one user record.
5. **users.append(user)** – Adds the newly registered user to the list.
6. **display_users()** – Displays all stored users.
7. **while True** – Keeps the program running until the user chooses to exit.
8. **if-elif-else** – Checks the user's menu choice and performs the selected operation.

Sample Output

1. Register User
2. Display Users

3. Exit

Enter your choice: 1

Enter your name: Raksha

Enter your email: raksha@gmail.com

User registered successfully!